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Dip. in Engg. (Sixth Semester)/ P. T. D. C. (Eighth Semester) EXAMINATION, Dec.-Jan., 2013-14

(Elect.)

SWITCHGEAR AND PROTECTION

Time : Three Hours] [*Maximum Marks : 100*

[*Minimum Pass Marks : 35*

Note : All questions are compulsory. Answers should be given to the point otherwise marks would be deducted. Solve any *two* parts from each question. Part (a) is compulsory.

1. (a) Explain what is meant by primary protection and backup protection. 10

Or

Explain the process of fault clearing with the help of a neat sketch.

(b) What is meant by 'Dead Zone' ? 5

Or

What is neutral earthing ?

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(c) What are the advantages of neutral earthing ? 5

Or

What is resonant grounding ?

2. (a) What is a large diverter ? What is basic principle of operation of a surge diverter ? 10

Or

What is surge absorber ? Write a short note on Ferranti surge absorber.

(b) Why are steep fronted surges more dangerous to power system equipment ? 5

Or

Explain cause and effect of overvoltage.

(c) Fill in the blanks : 5

(i) The most severe surges on the line are produced by ..... .

(ii) Arcing ground can be prevented by ..... .

(iii) A surge diverter should be located ..... to the apparatus to be protected.

(iv) In substation, the most commonly used type of arrester is ..... arrester.

(v) A 1/50 μs surge is ..... harmful than 3/50 μs surge. Assume the same peak value.

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3. (a) Describe the Merze price circulating current system for the protection of transformer. 10

Or

What is an IDMT characteristic ? Explain how this is achieved in practice.

(b) What is meant by 'directional feature' of a directional overcurrent relay ? Describe the construction, principle of operation. 10

Or

Explain clearly the basic principle of operation of a percentage differential relay. Explain the working of this type of relay :

(i) an internal fault.

(ii) a through fault.

(c) Determine the time of operation of a relay of rating 5 A, 2·2 sec. IDMT and having a relay setting of 125%. TMS = 0·6. It is connected to supply circuit through a C. T. 400/5 ratio. The fault current is 4000 A. 10

Or

Describe with a neat diagram, a circulating current protection scheme for a 3-phase, 1 MVA, 11 kV/400 V, delta-star

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transformer. If the current transformer have a nominal secondary current of 5 Amp, calculate their ratio.

4. (a) Draw the connection diagram of 3-phase protective system : 10

(i) A 3-phase transformer of 220/11000 line volts is connected in star/delta primary round.

(ii) A 3-phase transformer having line voltage ratio of 0·4 kV/11 kV is connected star-delta.

Or

Prove :

$$\theta = \frac{I_u \cos\delta - I_w \sin\delta}{n\, I_s}$$

Also explain polarity marking of current transformer.

(b) Polarity marking of potential transformer. 5

Or

Draw phasor diagram of potential transformer and explain it.

(c) Determine the VA output of a current transformer having a ratio of 100 : 5 and secondary resistance of 0·1 Ω (ohm). The resistance of connecting lead is 0·1 ohm and the relay burden is 5 VA. 5

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5. (a) What is an HRC fuse ? Compare an HRC fuse with a circuit breaker as an interrupting device. 10

Or

Explain the following terms :

(i) Pre-Arcing time

(ii) Arcing time

(iii) Cut-off

(b) Discuss the principle of operation of an air-blast circuit breaker. What are the advantages and disadvantages of using air as the arc quenching medium ? 10

Or

Write a short note on the rate of re-striking voltage indicating its importance in the arc extinction.

(c) (i) Distinguish between isolator and circuit breaker. 10

(ii) Current choping.

Or

(i) Circuit breaker rating.

(ii) Resistance switching.

6. (a) Discuss the important fault on an alternator. 10

Or

Discuss the protection of Induction motor.

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(b) Discuss the important fault of transformer. Explain Buchholz relay for transformer protection. 5

(c) Why is overload protection not necessary for alternator ? 5

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